How Far Can Labels Go? Benchmark Difficulty Is the Bottleneck of Cold-Start Predictive Evaluation

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Abstract

Predictive evaluation estimates whether an AI system will solve a benchmark item from a few observed outcomes. We study the benchmark-level cold start of the Predictive AI Evaluation Competition, where every test item comes from a benchmark absent from the training data and at most 31 outcomes are revealed per system–benchmark pair. Because each formative submission is scored on a new random sample, raw scores mix method quality with sample difficulty. We recover each pair's success rate from its budget-0 Brier score and calibration error and build a proxy that compares label-only predictors on 69 real test pairs and reproduces our submissions' scores within label noise. On this proxy, hierarchical Beta-Binomial shrinkage (HBS), which requires no training, improves on the official baseline by 0.028 Brier ALC and outperforms a Bayes-optimal predictor that uses only each pair's own labels. More than half of its remaining gap to an oracle comes from not knowing a new benchmark's difficulty. Item-level LLM difficulty estimates did not transfer to the test benchmarks, but averaging Qwen3.8-27B's ratings over a benchmark's items gives a prior that recovers two fifths of this component, a further 0.007. On a fresh test sample, it lowered the budget-0 Brier score from 0.224 to 0.190. HBS with this prior is our final submission.

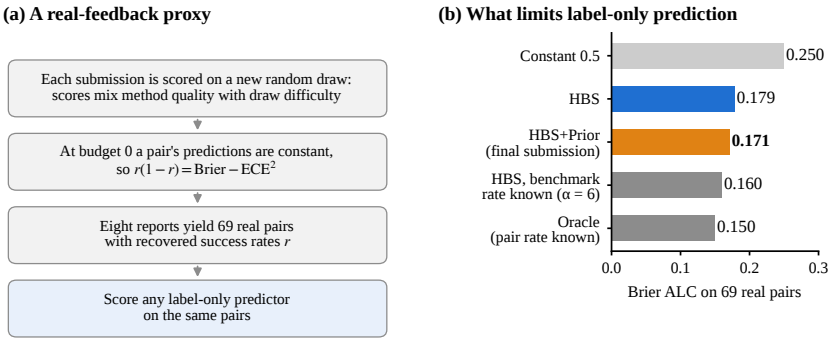


Figure 1: Overview. (a) Recovering each pair's success rate from formative feedback allows all label-only predictors to be scored on the same 69 real pairs. (b) Brier ALC on these pairs. Knowing each benchmark's success rate in advance ($\alpha = 6$) would remove most of the gap between HBS and the oracle; the LLM benchmark prior of our final submission recovers two fifths of that improvement.

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1 Introduction

Evaluating every AI system on every benchmark item is increasingly expensive, particularly for agentic benchmarks, in which a single attempt may involve a long tool-use trajectory. Predictive evaluation reduces this cost by estimating the probability that a system solves an item from attributes of the system and the item and from a few observed outcomes [Truong et al., 2025, Maia Polo et al., 2024]. Most existing methods calibrate item parameters on a population of systems that have already been evaluated on the same benchmark.

The Predictive AI Evaluation Competition (PAIEC) [Stanford AI Measurement Science, 2026] removes this assumption. Every test item comes from a benchmark absent from the public training pool, a predictor observes at most 31 outcomes per subject–benchmark pair, and predictions are scored by the Brier score averaged over label budgets, an area under the learning curve (Brier ALC). We call this setting *benchmark-level cold start*: before the first label arrives, nothing measured on the test benchmark is available, and afterwards too little is available to calibrate item parameters.

Developing methods for this setting also raises a measurement problem. Formative feedback for each submission is computed on a newly sampled set of pairs, and these samples differ widely in difficulty. Across the eight samples from which we build our proxy, an oracle that knows each pair's success rate scores between 0.099 and 0.183, whereas the methods we wished to compare differ by less than 0.01. Raw scores can therefore rank methods incorrectly. A variant of our method with a subject-ability adjustment scored 0.164 and HBS scored 0.188, yet on identical pairs the variant is worse by 0.026 (Section 6.1).

We make three contributions (Figure 1):

1. **The bottleneck is the benchmark prior.** More than half of the gap between a strong label-only predictor and an oracle comes from not knowing a new benchmark's difficulty before its labels arrive (Section 6.2). Item-level difficulty estimates from open LLMs did not transfer to the test benchmarks, but averaged over a benchmark's items, the ratings of a recent open model recover about two fifths of the missing prior (Sections 6.3 and 6.4).
2. **A real-feedback proxy.** When the budget-0 prediction is constant within a pair, the pair's success rate follows from its budget-0 Brier score and expected calibration error (ECE) (Proposition 1). Our eight reports yield 69 pairs on which label-only predictors can be compared directly, and the proxy reproduces our submissions' scores within label noise (Section 5).
3. **A strong label-only predictor.** Hierarchical Beta-Binomial shrinkage (HBS) pools labels across subjects and benchmarks and needs no training or pretrained model (Section 4). On the proxy it outperforms a Bayes-optimal predictor restricted to each pair's own labels, and no label-only variant we tested improves on it by more than 0.0004 (Section 6.1).

Our final submission combines HBS with the LLM benchmark prior (HBS+Prior).[1] Code to reproduce all results is available at `https://github.com/kay0601/how-far-can-labels-go`.

2 Related work

Efficient and predictive evaluation. Item response theory (IRT) [Lord, 1980] underlies most work on cheaper evaluation: tinyBenchmarks [Maia Polo et al., 2024] estimate accuracy from about 100 curated items, ATLAS [Li et al., 2026] selects items by Fisher information, and Truong et al. [2025] predict item difficulty from content. All need item parameters from systems already evaluated on the same benchmark, which benchmark-level cold start lacks. HBS instead estimates pair-level success rates.

Extending evaluation to new benchmarks. Habba et al. [2026] place new benchmarks on an existing IRT scale using anchor items answered by many models, which one or two subjects per test benchmark cannot provide. Observational scaling laws [Ruan et al., 2024] and BenchPress [Zeng and Papailiopoulos, 2026] predict a model's scores on held-out benchmarks from its other scores. Here, the benchmarks rather than the models are new.

[1] Codabench submission 978186. HBS alone, which needs neither a GPU nor a pretrained model, is submission 952533.

Predicting difficulty from content. Krsteski and Meyer [2026] predict task difficulty across 17 agentic benchmarks from reasoning-trace entropy, embeddings and metadata. Their leave-one-benchmark-out Spearman correlation of 0.225 is close to our 0.24 to 0.28 (Section 6.3), and they describe transfer to unseen benchmarks as open. Zhou et al. [2025] annotate 18 demand rubrics per item with GPT-4o and train an assessor for each AI system, reaching an AUROC of 0.747 on held-out benchmarks, but need a proprietary annotator and training data for each system, whereas the test subjects are new. PRECOG [Park et al., 2025] forecasts task-level performance from task descriptions alone.

Shrinkage and probabilistic forecasts. HBS is an empirical-Bayes estimator in the tradition of Efron and Morris [1975]. Predictions are scored with the Brier score [Brier, 1950] and expected calibration error [Guo et al., 2017], and our proxy builds on the single-bin case of Murphy's decomposition [Murphy, 1973].

Concurrent work. Another PAIEC entry [aghasalim, 2026] uses a shrunk logistic predictor and compares versions by regressing each pair's Brier ALC on its budget-0 Brier score over 42 submissions, whereas our proxy compares methods on identical pairs. It also finds that item-text features and sentence embeddings do not transfer to held-out benchmarks.

3 Problem setup

Task. A *subject* s is an AI system together with its evaluation settings (model, provider, dates, harness, reasoning effort). An *item* i belongs to a benchmark $b(i)$ and is described by its text, structured features (`item_features`), interactors and an anonymous benchmark identifier. The outcome $y_{si} \in \{0, 1\}$ records whether s solved i. A *pair* $j = (s, b)$ is a subject–benchmark combination with at least 80 items, split evenly into acquisition items A_j and evaluation items E_j; its *success rate* is $r_j = |E_j|^{-1} \sum_{i \in E_j} y_{si}$. A *draw* D is the set of pairs sampled to score one submission. At label budget $t \in \mathcal{T} = \{0, 1, 3, 7, 15, 31\}$, the outcomes of up to t items from A_j have been revealed for every pair $j \in D$; we write $\mathcal{L}_t$ for all labels revealed in the draw. A predictor $f(s, i; \mathcal{L}_t) \in [0, 1]$ is called for every evaluation item at every budget, and every call receives all of $\mathcal{L}_t$. By default, labels are acquired in random order, but a submission may supply its own acquisition policy.

Metric. At budget t, pair j is scored by the Brier score of its evaluation items, and Brier ALC averages a weighted sum over budgets across the pairs of a draw:

$$\mathrm{Brier}_j(t) = \frac{1}{|E_j|} \sum_{i \in E_j} \big(f(s, i; \mathcal{L}_t) - y_{si}\big)^2, \qquad \mathrm{ALC}(f; D) = \frac{1}{|D|} \sum_{j \in D} \sum_{t \in \mathcal{T}} w_t \, \mathrm{Brier}_j(t), \tag{1}$$

with $w_0 = w_{31} = 0.1$ and $w_1 = w_3 = w_7 = w_{15} = 0.2$, so that the weights sum to one. Lower is better. For each pair and budget, formative feedback reports $\mathrm{Brier}_j(t)$ and the expected calibration error over ten equal-width bins,

$$\mathrm{ECE}_j(t) = \sum_{m=1}^{10} \frac{|E_{jm}|}{|E_j|} \Big| \frac{1}{|E_{jm}|} \sum_{i \in E_{jm}} \big(f(s, i; \mathcal{L}_t) - y_{si}\big) \Big|, \tag{2}$$

where $E_{jm} \subseteq E_j$ contains the items whose prediction falls in bin m.

Data. Benchmarks were assigned at random to disjoint training and test pools before curation. We use the public training pool of measurement-db [AIMS Foundations, 2026], four benchmarks of which have several subjects per item (Table 4 in Appendix C). Each formative submission is scored on a draw of at most 1,000 subject–item pairs. The eight draws from which we build the proxy contain 8 or 9 pairs each, mostly with one subject per benchmark, from seven test benchmarks in total. The final ranking uses a common hidden test subset.

Offline simulator. Methods that use item text, and those developed before the proxy, are evaluated on 400 draws simulated from the four multi-subject training benchmarks like test draws: about eight pairs per draw, mostly one subject per benchmark, 88 to 230 items per pair and random acquisition. Methods that use item text are selected leave-one-benchmark-out (LOBO).

4 Hierarchical Beta-Binomial shrinkage

HBS estimates a success rate for each pair by shrinking every level of a three-level hierarchy (global, benchmark, pair) toward the level above it, and then applies two small item-level corrections. HBS computes every quantity from $\mathcal{L}_t$ within the call and keeps no state between calls. Apart from its hyperparameters, nothing is fitted to training data.

Pair estimate. For a target pair (s, b), let k_{sb} and n_{sb} be the subject's successes and labels on b in $\mathcal{L}_t$, let $K_{-s,b}$ and $N_{-s,b}$ be those of all other subjects on b, and let K_{-b} and N_{-b} be those on all other benchmarks. HBS computes

$$g_b = \frac{K_{-b} + \kappa_0\,\mu_0}{N_{-b} + \kappa_0}, \qquad \pi_{sb} = \frac{K_{-s,b} + \kappa_1\,g_b}{N_{-s,b} + \kappa_1}, \qquad \bar{p}_{sb} = \frac{k_{sb} + \alpha\,\pi_{sb}}{n_{sb} + \alpha}. \tag{3}$$

Each step is the posterior mean of a Beta-Binomial model whose prior has its mean at the level above and a strength of κ_0, κ_1 or α pseudo-labels. The global rate g_b is shrunk strongly toward μ_0 because success rates differ widely across benchmarks, so labels on other benchmarks say little about b. The benchmark rate π_{sb} excludes the target subject's own labels, which already enter $\bar{p}_{sb}$ and would otherwise be counted twice. At budget 0, $\mathcal{L}_0$ is empty and HBS predicts μ_0 for every item.

Item and group corrections. Let $R(i)$ be the labels that other subjects received on the same item i, and $G(i)$ the labels, from any subject including s, on items of the same benchmark that share i's feature group, given by the non-numeric fields of `item_features` (Appendix C), which identify, for example, a contest, website, paper or programming language. For a multiset S of labels (s', i', y), let $\Delta(S) = \sum_{(s',i',y)\in S}(y - \bar{p}_{s'b(i')})$ be the summed residual of each label against its own pair's estimate. The prediction is

$$\hat{p}_{si} = \mathrm{clip}_{[\epsilon,\,1-\epsilon]}\Big(\bar{p}_{sb} + \gamma_I\,\frac{\Delta(R(i))}{|R(i)| + \kappa_I} + \gamma_G\,\frac{\Delta(G(i))}{|G(i)| + \kappa_G}\Big). \tag{4}$$

The item correction is rarely active in test draws, which seldom contain two subjects on the same benchmark. The group correction transfers within-benchmark difficulty structure, such as a hard contest, once its labels arrive. Because $G(i)$ includes the target pair's own labels, it also weakens the shrinkage of $\bar{p}_{sb}$ when a benchmark's items share one group, and the hyperparameters were tuned with this behavior.

Hyperparameters. We use $\mu_0 = 0.35$, $\kappa_0 = 100$, $\kappa_1 = \alpha = 3$, $\kappa_I = \gamma_I = 1$, $\kappa_G = 5$, $\gamma_G = 0.75$ and $\epsilon = 0.01$ (Table 5 in Appendix C). They were chosen with the offline simulator (Section 3) and two early feedback reports, and re-tuning them on the proxy does not improve the score (Section 6.1). HBS is about 170 lines of standard-library Python.

LLM benchmark prior (HBS+Prior). Our final submission replaces the constant μ_0 with a benchmark-specific prior estimated from item text. Each item to be predicted is rated by Qwen3.8-27B [Qwen Team, 2026b], quantized to 4 bits [Dettmers et al., 2023], with an anchored prompt that shows twelve items from the four multi-subject training benchmarks together with their observed success rates (Appendix B). The rating is $\rho_i = (\mathbb{E}[d] + 0.5)/10$, where $\mathbb{E}[d]$ is the expected digit under the model's next-token distribution over the digits 0 to 9. With I_b the items rated so far on benchmark b, the prior

$$\mu_b = \frac{\sum_{i\in I_b}\rho_i + K_0\,\mu_0}{|I_b| + K_0}, \qquad K_0 = 5, \tag{5}$$

replaces μ_0 in the global rate g_b of Equation (3). All other components are unchanged, so labels on other benchmarks still dilute μ_b through g_b, mainly at large budgets, where the pair's own labels dominate. Unlike HBS, HBS+Prior keeps state. It caches ratings across calls, budgets and subjects, so μ_b averages all rated items of the benchmark from budget 1 onward and grows item by item at budget 0, where its predictions therefore vary within a pair (Appendix C).

5 Evaluating methods with formative feedback

To compare methods on the same pairs, we recover the success rate of every pair in our feedback reports and build a proxy on these rates.

5.1 Recovering pair success rates

Proposition 1. *If the budget-0 prediction is a constant p_j on the evaluation items of pair j, then* $\mathrm{Brier}_j(0) = r_j(1-r_j) + (p_j - r_j)^2$ *and* $\mathrm{ECE}_j(0) = |p_j - r_j|$. *Consequently,*

$$r_j(1-r_j) = \mathrm{Brier}_j(0) - \mathrm{ECE}_j(0)^2. \tag{6}$$

Proof. Because $y_{si}^2 = y_{si}$, $\mathrm{Brier}_j(0) = |E_j|^{-1} \sum_{i \in E_j} (p_j^2 - 2p_j y_{si} + y_{si}) = p_j^2 - 2p_j r_j + r_j$, which equals $r_j(1-r_j) + (p_j - r_j)^2$. All predictions fall into the same bin, so Equation (2) reduces to $|p_j - r_j|$. □

The condition holds for HBS, and for any predictor whose budget-0 output depends only on the subject and the benchmark, because no labels are available at budget 0. Equation (6) determines r_j up to the reflection $r_j \leftrightarrow 1 - r_j$. When $p_j \neq 1/2$ is known, $\mathrm{Brier}_j(0) = p_j^2 + (1 - 2p_j)\, r_j$ gives r_j directly. The prediction is unknown for 43 pairs: 34 from four early submissions whose code we did not retain, and 9 from HBS+Ability, whose prediction is constant within a pair but depends on subject metadata that the logs anonymize. For these, we choose the root closer to the success rates of the same benchmark in the reports whose prediction is known. On the 26 pairs with known roots, from the reports of the smoothed mean, HBS and HBS+ItemLLM, this rule selects the correct root in 25 cases (Appendix E).

Oracle limit of a draw. A predictor that knows every r_j and predicts it at every budget scores $\mathrm{Brier}_j(t) = r_j(1-r_j)$, so by Equation (1) its Brier ALC is

$$\bar{v}(D) = \frac{1}{|D|} \sum_{j \in D} r_j(1-r_j). \tag{7}$$

By Proposition 1, no predictor that is constant within each pair can score below $\bar{v}(D)$. We call $\bar{v}(D)$ the *oracle limit* of the draw. It measures how difficult a draw is, independently of any method. Across our eight reports it ranges from 0.099 to 0.183 (Figure 3 in Appendix A), and the two best raw scores were obtained on the two easiest draws.

5.2 The proxy

The proxy scores a label-only predictor f on the 69 recovered pairs. For each report, we draw label sequences $y_{j1}, \ldots, y_{j31} \sim \mathrm{Bernoulli}(r_j)$ independently for its pairs, reveal the first t labels of every pair at budget t, and query f with item inputs that carry the subject and the benchmark identifier but no content or features. Since the outcomes on E_j average to r_j, the Brier score of a prediction $p_j(t)$ is exactly $r_j(1-r_j) + (p_j(t) - r_j)^2$, and averaging over label sequences gives f's expected Brier ALC on the report. The pairs of a report are simulated jointly, so pooling across pairs behaves as it does on the real draw. Parameters are selected leave-one-report-out, and 95% intervals are obtained by resampling whole reports.

Validity. For the three submissions whose code we retained (the smoothed mean, HBS+Ability and HBS; Appendix A), Table 1 compares the real score with the proxy's expectation for the same predictor on the same pairs, and with the spread due to label noise alone, that is, to which outcomes happen to be revealed. Every real score lies below its expectation, by 0.006 to 0.010, but within 1.3 standard deviations. Disabling the item and group corrections and ignoring that labels are drawn without replacement from a finite pool each account for about 0.001 of this difference (Appendix E). At the pair level, 24 of the 25 pairs with label noise lie within two standard deviations. We therefore use the proxy as our primary criterion for label-only predictors.

5.3 Measuring an LLM prior on the test benchmarks

The proof of Proposition 1 applies at any budget at which predictions are constant within each pair, in which case $\mathrm{ECE}_j(t) = |p_j(t) - r_j|$. We used this in one measurement submission, not intended as a final entry, to estimate the LLM prior of Equation (5) on the test benchmarks. It predicted the constant 0.35 at budget 0, which identifies r_j; the prior μ_b at budget 1, which yields $|\mu_b - r_j|$; and μ_b+0.1 at budget 3, which determines on which side of r_j the prior lies. In simulation, this procedure

Table 1: Real Brier ALC and the proxy's expectation for the same predictor on the same pairs; SD is the spread due to label noise alone, and differences are computed before rounding. HBS+Ability's rates are inferred (Section 5).

Submission	Real	Proxy	SD	Real − proxy	z
Smoothed mean	0.1829	0.1899	0.0069	-0.0070	-1.0
HBS+Ability	0.1639	0.1741	0.0076	-0.0103	-1.3
HBS	0.1883	0.1946	0.0077	-0.0063	-0.8

Table 2: Label-only alternatives to HBS. Positive changes in Brier ALC are worse; intervals are 95% intervals from resampling whole reports with the inferred roots held fixed, and the single-draw HBS+Ability row gives the SD of label noise. HBS scores 0.179 on the proxy.

Alternative to HBS	Selection	Change in Brier ALC
Real-feedback proxy, 69 pairs		
Official empirical-mean baseline	reference	$+0.0282$ $[+0.0234, +0.0324]$
Smoothed mean	fixed	$+0.0028$ $[+0.0013, +0.0043]$
Benchmark pooling removed ($\kappa_1 \to \infty$)	fixed	$+0.0030$ $[+0.0015, +0.0045]$
Bayes-optimal own-label predictor	reference	$+0.0024$ $[+0.0014, +0.0034]$
Mixture prior with a near-zero component	leave-one-report-out	$+0.0004$ $[-0.0004, +0.0008]$
Re-tuned μ_0, α, κ_1	leave-one-report-out	$+0.0006$ $[-0.0008, +0.0015]$
HBS+Ability, on its own draw	fixed	$+0.0265$ (SD 0.0045)
Offline simulator, 400 draws		
Stratified label acquisition	fixed	-0.0004 (SE 0.0001)
Group correction removed	fixed	$+0.0011$ (SE 0.0001)

recovers μ_b exactly, and in the real run the budget-1 scores of every pair were consistent with a constant prediction. Because benchmark identifiers are fixed across submissions, the recovered priors allow the proxy to score HBS+Prior on all 69 pairs.

Use of formative feedback. The proxy and the measurement submission rely only on our own formative feedback, which the competition provides to guide development. We used the recovered rates and priors only to compare methods offline. No recovered value is written into any submitted program, we made no attempt to identify the subjects or benchmarks behind the anonymous identifiers, and we do not release these values.

6 Experiments

6.1 Label-only prediction

On the 69 proxy pairs, HBS scores 0.179, compared with 0.207 for the organizers' empirical-mean baseline, 0.250 for a constant prediction of 0.5 and 0.150 for the oracle (Table 2). Pooling the labels of other subjects on the same benchmark matters. Removing it costs 0.0030, as much as replacing HBS with the simpler smoothed mean (0.0028). A Bayes-optimal predictor that knows the distribution of the other pairs' success rates but sees only the target pair's labels is worse than HBS by 0.0024, although estimating that distribution on the same reports favors it. Neither a mixture prior with a near-zero component nor re-tuning μ_0, α and κ_1 on held-out reports improves on HBS, and stratified label acquisition gains only 0.0004 in simulation. These conclusions also hold on the six reports that did not inform the design of HBS and on the 26 pairs whose root is known (Appendix E).

Subject-ability adjustment. Information about the subjects did not help. HBS+Ability shifts the benchmark prior in logit space by a subject-ability score estimated on the training data. It helped in simulation, but on real feedback it raised the budget-0 Brier score to 0.326, worse than predicting 0.5, and lost 0.026 to HBS on its own draw. The training pool contains many older systems, whereas the test subjects are recent systems evaluated on harder benchmarks.

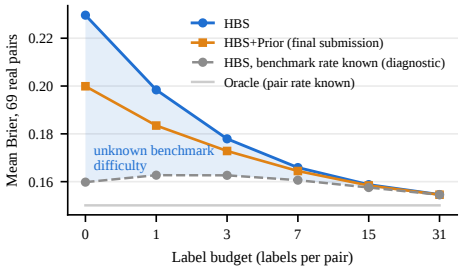


Figure 2: Mean Brier score per label budget on the 69 real pairs. Knowing each benchmark's rate ($\alpha = 6$) removes most of the error of HBS at low budgets; the LLM prior recovers part of it.

6.2 Where the remaining error lies

On the 69 pairs, HBS is 0.029 above the oracle. To locate this gap, we gave HBS each benchmark's mean success rate in advance, estimated from the other reports, in place of μ_0. With the pair-prior strength of HBS ($\alpha = 3$), this lowers its score to 0.162 and closes 58% of the gap (95% interval 54 to 61%). With $\alpha = 6$, which suits an accurate prior, it closes 65% (60 to 69%). On the 26 pairs with known roots, the two settings close 54% and 56%. The loss is concentrated at budgets 0 to 3, where HBS relies on its global prior, and vanishes by budget 31 (Figure 2). The remainder reflects differences between subjects on the same benchmark. This diagnostic uses rates recovered from test feedback and could not be part of a submission.

6.3 Can item text close the gap?

Before any labels arrive, item text is the only information available about a new benchmark. We evaluated open LLMs that fit on the competition's hardware, holding out one of the four multi-subject training benchmarks at a time and using each item's success rate averaged over subjects as the target (prompts in Appendix B; further details in Appendices C and F).

Item difficulty within a benchmark. We prompted Qwen3-8B and Qwen3-32B [Yang et al., 2025] to rate 5,322 training items and fitted a ridge-regression probe on the hidden state of the last prompt token. On the held-out benchmark, the probe ranks items with an average Spearman correlation of 0.24 (8B) and 0.28 (32B), compared with -0.03 for bge-small sentence embeddings [Xiao et al., 2024]. In the simulator, the probe scores improved Brier ALC by 0.003 and 0.005, respectively.

Benchmark difficulty. Without examples, both Qwen3 models rated every training benchmark between 6.8 and 7.7 out of 9 on average, although the benchmarks' mean success rates range from 0.22 to 0.62. With nine anchor items from the other benchmarks and their observed success rates in the prompt, only Qwen3.8-27B, the most recent of the four models, produced a benchmark-level prior better than the constant of HBS, with a budget-0 Brier score of 0.217 against 0.233. Qwen3-8B, Qwen3.5-9B [Qwen Team, 2026a] and the larger but older Qwen3-32B did not (Table 10). A single rating is noisy, but a predictor can average the ratings of all items it is asked about on a benchmark. Four training benchmarks are too few to validate this, so we evaluated it on the test benchmarks (Section 6.4).

Item-level shift on real feedback. To test the item-level signal on real data, we submitted HBS+ItemLLM, which shifts the HBS estimate in logit space by $\gamma(z_i - c)$, where z_i is the item's standardized Qwen3-32B probe score, c is the mean score of the labeled items of the same benchmark and the target item, and $\gamma = 0.5$. It improved Brier ALC by 0.005 in the simulator, with γ selected leave-one-benchmark-out, and ran as intended on the platform, yet it scored 0.157 on its real draw, where the proxy expects 0.131 for HBS, a difference of 3.8 standard deviations of label noise. The loss is concentrated at budgets 1 to 7 and vanishes by budget 15 (Figure 5b), so it stems from centering on one to three labeled items, where c depends on one or two scores and shifts a

whole pair. Its size suggests that the scores, standardized with training statistics, spread more widely on the test items, which four training benchmarks could not reveal.

6.4 A benchmark prior from an LLM

The measurement submission (Section 5.3) recovered the prior of all seven test benchmarks. The priors order the benchmarks' mean success rates across our reports with Kendall's $\tau = 0.62$ and are closer to them than the constant 0.35 for five of the seven, but they are compressed toward the middle (Appendix E). Used from budget 1, the prior improves on HBS by 0.004. Used from budget 0, where it accumulates item by item, it improves on HBS by 0.007 (95% interval 0.006 to 0.009; Table 7 in Appendix E), mostly at budgets 0 to 3 (Figure 2). This is 44% of what knowing each benchmark's rate gains at the same α, and 39% with $\alpha = 6$. On the 26 pairs with known roots, the improvement is 0.008 and holds in each of their three reports. Removing any one benchmark leaves 0.006 to 0.009 (jackknife standard error over benchmarks 0.003), and the improvement changes little with K_0, the order in which items are rated, or a redraw of each prior within its sampling error (Appendix E). These estimates are in-sample with respect to the test pool, because the same seven benchmarks supplied the priors and make up the 69 pairs. On the training benchmarks, whose rates lie closer to 0.35, the same procedure gained only about 0.002 in simulation, which suggests that a benchmark prior helps most when rates are extreme, as in the test pool.

The formative run of the final submission is a check on a fresh draw (Table 9 in Appendix E). At budget 0, which involves no labels and hence no label noise, its Brier score was 0.190, against 0.224 for the HBS constant on the same pairs. Over all budgets it scored 0.174, between the proxy's expectations for HBS (0.180) and HBS+Prior (0.171) and within label noise of both.

7 Discussion and limitations

Benchmark-level cold start is limited by information. How difficult a benchmark is for current systems depends on grading strictness, the harness and environment, and recent progress in model capabilities, none of which is stated in its items. Success rates in the test pool cluster by benchmark, so the first few labels are highly informative, whereas a text-based prior helps only if it reflects current capabilities. Only the most recent LLM we rated with anchors helped, not the larger Qwen3-32B, and only when its ratings were averaged over a benchmark's items. Current LLMs thus appear to know the approximate difficulty of a kind of task better than that of an individual item.

Simulation can mislead. Two changes that helped on the public training pool hurt on real feedback, and both relied on statistics learned across that pool. The subject-ability adjustment assumed its older population of systems, and the item-level shift assumed that difficulty signals from four benchmarks transfer. Leave-one-benchmark-out validation on so few benchmarks cannot detect such shifts, so we adopted a change only if it also held on real feedback.

Limitations. The proxy omits item and group structure, treats labels as independent Bernoulli draws, is pessimistic by 0.006 to 0.010 (Table 1) and cannot evaluate methods that use item text. Its intervals resample eight reports and hold the inferred roots and the seven benchmarks fixed. For 43 of its 69 pairs, the root is inferred by a rule that is correct on 25 of 26 known pairs. The comparisons with the own-label predictor and with HBS without pooling, and the share due to benchmark difficulty, depend on these roots (Appendix E). The 69 pairs come from seven test benchmarks, and the hidden final test subset may differ. The verdict on the item-level shift is based on one real draw, although five of its nine pairs showed the loss. The LLM prior was measured once, on the seven test benchmarks. Its value elsewhere depends on how well the model judges other benchmarks, and without a GPU the final submission falls back to HBS. We tested open models of at most 32 billion parameters, and larger models or proprietary APIs may do better.

8 Conclusion

In benchmark-level cold start, the main limitation is missing information about the new benchmark rather than an insufficiently elaborate predictor. No label-only alternative we tested improves on hierarchical shrinkage by more than 0.0004, and more than half of its remaining error comes from not

knowing an unseen benchmark's difficulty. Averaging a recent open LLM's ratings over a benchmark's items supplies about two fifths of this missing prior, and our final submission uses it. A reliable judge of benchmark difficulty would be worth up to 0.019 Brier ALC on our pairs.

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A Submissions

Table 3 lists the formative submissions referred to in this paper. In Figure 3, *Early 1* to *Early 4* denote the four early submissions in the order listed. Their predictors are not evaluated in this paper, but their feedback reports contribute 34 pairs to the proxy.

Table 3: Formative submissions and their Codabench identifiers.

Method	Description	Codabench ID
Early submissions	Our first predictors, made before the methods in this paper; code not retained	928061, 930136, 930165, 931142
Smoothed mean	Pair's label mean shrunk ($\alpha = 5$) toward the benchmark's label mean, itself shrunk toward 0.35 with strength 30; with the item correction	951748
HBS+Ability	HBS with a prior mean of 0.4 and a subject-ability shift estimated on the training data	951853
HBS	Section 4; label-only	952533
HBS+ItemLLM	HBS with an item-level shift from Qwen3-32B representations (Section 6.3)	960385
Measurement	Measures the LLM benchmark prior on the test benchmarks (Section 5.3); not a candidate for the final ranking	978024
HBS+Prior	HBS with the LLM benchmark prior from budget 0 (Section 4); final submission	978186

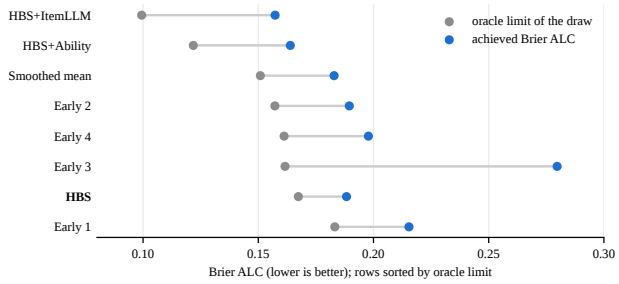


Figure 3: Oracle limit and achieved Brier ALC of the eight formative submissions used to build the proxy. The oracle limit measures the difficulty of each draw, independently of the method.

B Prompts

All prompts were formatted with the model's chat template, with thinking disabled, followed by the prefilled answer prefix "`Rating:␣`", so that the next token is the rating. We read the probabilities of the digit tokens 0 to 9 at that position and renormalize them. The rating is the expected digit $\mathbb{E}[d]$; for the anchored prompt, it is mapped to a success rate $(\mathbb{E}[d] + 0.5)/10$. The item-difficulty probe uses the hidden states of the zero-shot prompt rather than its rating. `{content}` denotes the item text. A text longer than its limit keeps its first h tokens followed by a window taken from its end (h = 384 of 2,048 tokens for zero-shot ratings, 64 of 256 for anchors and 192 of 1,024 for targets); if a `TODO` marker of a code-completion item lies before that window, the window instead ends up to 512 tokens after the marker.

Zero-shot rating (item difficulty). Item texts were truncated to 2,048 tokens.

> Below is a task from an AI evaluation benchmark. A capable AI system (a frontier language model, possibly acting as an agent with tools) will attempt it, and the attempt is graded pass/fail.
>
> `<task>`
> `{content}`
> `</task>`

> Do not attempt the task yourself. How likely is a strong AI system to solve this task correctly? Answer with a single digit from 0 to 9, where 0 means it almost certainly fails and 9 means it almost certainly succeeds.

Anchored rating (benchmark difficulty). The introduction is followed by the examples in a fixed random order and then by the question: nine examples from the other three benchmarks in the experiments of Section 6.3, and twelve from all four benchmarks in HBS+Prior. The anchors of a benchmark are its items at the 20th, 50th and 80th percentiles of success rate among those with at least five subjects; `{pct}` is the success rate as a rounded percentage and `{digit}` is $\min(9, \lfloor 10 \times \text{rate} \rfloor)$. In Section 6.3, the targets are 120 random items per benchmark with at least three subjects (seed 0); HBS+Prior rates every item it is asked about.

> You will estimate how often AI systems succeed on a task from an AI evaluation benchmark. The AI systems are language models, often acting as agents with tools, and each attempt is graded pass/fail.
>
> For reference, here are tasks from other benchmarks with the share of AI systems that solved them. A digit d means d*10% to (d+1)*10%.
>
> `<example>`
> `<task>`
> `{content}`
> `</task>`
> Share of AI systems that solved it: `{pct}`% (digit `{digit}`)
> `</example>`
>
> [... eight further examples (eleven in HBS+Prior) ...]
>
> Now a new task, from a different benchmark:
>
> `<task>`
> `{content}`
> `</task>`
>
> Do not attempt the task yourself. What share of AI systems solve this task? Answer with a single digit from 0 to 9 as defined above.

C Reproducibility and implementation details

Code and data. Code is available at https://github.com/kay0601/how-far-can-labels-go; its README lists the command that produces each table and figure. The training data are revision bc8204d of measurement-db, obtained with the official preparation script; `matharena`, which that script excludes, is downloaded separately at the same revision. Formative feedback is private to each team and is not released; the proxy scripts read a team's own scoring logs. The official preparation script excludes `matharena` because about 60 of its responses are not binary; we drop those responses and keep the rest. `mmdocrag` is not used because its scores are continuous, and `swe_rebench`, with one subject per item, is used for training only.

Feature groups. `item_features` is a list of `key=value` fields separated by semicolons. The group of an item consists of its benchmark and its fields whose value is not a number, except `prompt_source`; an item without such fields has no group and no group correction.

HBS+Prior. The platform restarts the prediction workers at each budget, so ratings are cached on disk and each item is rated only once. Each benchmark's sum and count of ratings are shared by all subjects on it, so at budget 0 the prior used for an item depends on how many items of its benchmark, including other subjects' items, were rated before it. Rating one item takes about 3 seconds on an RTX 3090, well within the platform's limit of 120 seconds per call; the 4-bit model needs about 18 GB of GPU memory and is downloaded by the platform from the model list of the submission. If the model cannot be loaded, the submission falls back to HBS.

LLM experiments. The ridge probe of Section 6.3 uses a regularization strength of 10^4 and the hidden state at the best of three depths: layer 27 of 36 for Qwen3-8B and the final layer, 64, for Qwen3-32B. Qwen3-32B and Qwen3.8-27B were loaded in 4-bit precision, and Qwen3-8B and Qwen3.5-9B in bfloat16.

Table 4: Public training benchmarks (measurement-db, revision bc8204d). Only benchmarks with several subjects per item can be used to evaluate methods offline.

Benchmark	Items	Subjects per item (median)	Use
`multi_swebench`	2,126	27	training and offline evaluation
`matharena`	1,751	9	training and offline evaluation
`real_webagents`	233	12	training and offline evaluation
`researchcodebench`	212	31	training and offline evaluation
`swe_rebench`	6,306	1	training only

Table 5: Hyperparameters of HBS, with their names in the released code.

Symbol	Code name	Role	Value
μ_0	`PRIOR_MEAN`	mean of the global prior	0.35
κ_0	`GLOBAL_K`	strength of the global prior	100
κ_1	`BENCH_K`	strength of the benchmark prior	3
α	`ALPHA`	strength of the pair prior	3
κ_I, γ_I	`ITEM_K`, `ITEM_GAMMA`	shrinkage and scale of the item correction	1, 1
κ_G, γ_G	`GROUP_K`, `GROUP_GAMMA`	shrinkage and scale of the group correction	5, 0.75
ϵ	`CLIP`	clipping margin	0.01

Compute. One simulator configuration (400 draws) takes about 10 seconds on a CPU. The LLM experiments ran on a single RTX 3090 (24 GB): extracting Qwen3-8B representations for 5,322 items took 12 minutes and Qwen3-32B representations about 50 minutes; the anchored ratings took 10 minutes for Qwen3.5-9B, 22 minutes for Qwen3-32B and 28 minutes for Qwen3.8-27B. The formative run of the measurement submission, with 491 evaluation items, completed in under 40 minutes. Software: Python 3.13 for HBS and the simulator; Python 3.10 with PyTorch 2.7.1, Transformers 5.2.0 and bitsandbytes 0.50.2 for the LLM experiments.

D Disclosures

- Training data: only the public training pool of measurement-db (CC BY-SA 4.0, gated access under its terms of use), as listed in Table 4. We do not redistribute the data. No external data and no contributions from other teams were used.
- Pretrained models: BAAI/bge-small-en-v1.5 (MIT license) and Qwen/Qwen3-8B, Qwen/Qwen3-32B, Qwen/Qwen3.5-9B and Qwen/Qwen3.8-27B (Apache 2.0). The final submission (HBS+Prior) loads Qwen3.8-27B at run time; HBS uses no pretrained model; HBS+ItemLLM loaded Qwen3-32B.
- Coding assistant: we used Claude Code (Anthropic) for implementation, analysis and drafting this report.
- Broader impact: predictive evaluation can reduce the compute spent on evaluating AI systems. Its main risk is that predicted success rates are trusted in place of measured ones; our results show that predictions are least reliable on unseen benchmarks before labels arrive, which is where such trust would be most tempting. Scores on small random samples also mix the quality of a method with the difficulty of the sample, which matters for anyone who ranks methods on such samples; reporting each draw's oracle limit next to its score, which any team can compute from its own feedback, would separate the two.

E Robustness of the proxy

Subsets. The 26 pairs with known roots come from the reports of the smoothed mean, HBS and HBS+ItemLLM, which predict 0.35 at budget 0. The 26 pairs of Table 1 and Figure 4 come from the reports of the three submissions whose code we retained, the smoothed mean, HBS+Ability and HBS; for HBS+Ability, the proxy uses the inferred budget-0 predictions, through an ability table that reproduces them.

Pair-level validity. Figure 4 compares each pair's real Brier ALC with the proxy's expectation for the three submissions of Table 1, as a z-score under label noise alone. Of the 25 pairs with label noise, 24 lie within two standard deviations, and the mean z of -0.35 reflects the proxy's pessimism. The exception ($z = 3.6$, on the draw of HBS) is a pair whose real Brier score rose from 0.27 at budget 0 to 0.34 at budgets 3 to 15. This is consistent with an acquisition half whose success rate differs from that of the evaluation half, which the proxy does not model because it draws labels from the evaluation rate. The one pair without label noise matches its expectation to within 10^{-6}.

Pessimism of the proxy. Drawing each pair's labels without replacement from a pool of $|E_j|$ items with the pair's success rate, instead of independently, lowers the proxy's expectation by 0.0008 to 0.0009 for the three submissions of Table 1, and the item and group corrections, which the proxy disables, are worth about 0.001 in the simulator (Table 2). These two effects explain about 0.002 of the 0.006 to 0.010 by which the real scores fall below the proxy. The rest is within label noise, although its consistent sign suggests a further difference between real and simulated label sequences, such as acquisition halves whose rates differ from those of the evaluation halves.

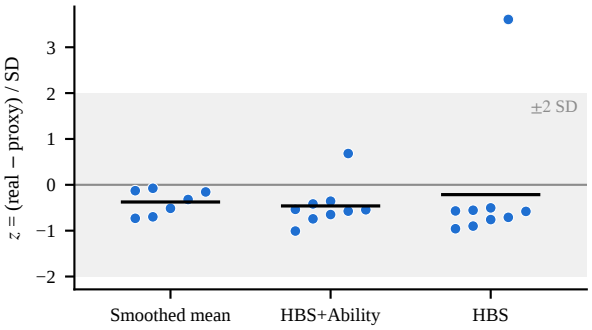


Figure 4: Real minus proxy-expected Brier ALC of each pair with label noise, in standard deviations of label noise, for the three submissions of Table 1. Bars mark the mean of each submission; the shaded band is ± 2 standard deviations.

Reports that did not inform the design. Because the reports of the smoothed mean and HBS+Ability informed the design of HBS, we repeated the comparisons of Table 2 with the official baseline, the smoothed mean, the ablation and the own-label predictor on the other six reports: every difference keeps its sign and changes by at most 0.001.

Choice of roots. For the 43 pairs whose budget-0 prediction is unknown, from the four early reports and the report of HBS+Ability, we chose between r_j and $1 - r_j$ using the rates of the same benchmark in the reports whose prediction is known. Applied to the 26 pairs whose root is known, with references from the other known reports, this rule picks the correct root for 25 and lands within 0.1 of it for all 26; the miss is a pair near 0.5, where the two roots nearly coincide. Table 6 repeats the main comparisons of Table 2 on the 26 known pairs alone, with all 43 inferred roots flipped, and with each inferred root flipped at random with probability 0.20, the upper end of the 95% interval for the rule's error rate (40 assignments, 300 label sequences each). The advantages of HBS over the official baseline and the smoothed mean hold in every case. The advantage over HBS without benchmark pooling holds on the known pairs and with all roots flipped, but reverses in some random assignments. The comparison with the own-label predictor and the share of the gap due to benchmark difficulty hold on the known pairs, where the share is 56%, but depend on the inferred roots: both reverse when all 43 roots are flipped, which would contradict the rates of the same benchmarks in other reports, and both become uncertain under random flips. The intervals of Tables 2 and 7 hold the inferred roots fixed. Independent random flips are a deliberately pessimistic test: they break the clustering of rates by benchmark that the rule relies on, whereas the rule's error is most likely for rates near 0.5, where a flip changes little.

Benchmark-difficulty diagnostic. The diagnostic replaces μ_0 by the benchmark's mean rate in the other reports. With $\alpha = 3$, it scores 0.162 and closes 58% of the gap between HBS and the

oracle (95% interval 54 to 61%); fixing g_b at that mean instead ($\kappa_0 \to \infty$) closes 59%. With $\alpha = 6$, used in Figures 1 and 2, it closes 65%. On the 26 known pairs, the shares are 54% and 56%.

Table 6: Change in Brier ALC of each alternative relative to HBS, and the share of HBS's gap to the oracle closed by knowing each benchmark's rate, under different choices of the 43 inferred roots.

Roots	Baseline	Smoothed mean	No pooling	Own-label	Share
As inferred, 69 pairs (Table 2)	$+0.028$	$+0.003$	$+0.003$	$+0.002$	0.65
26 known pairs only	$+0.024$	$+0.005$	$+0.005$	$+0.003$	0.56
All 43 flipped	$+0.017$	$+0.010$	$+0.006$	-0.004	-0.12
Random flips, minimum over 40	$+0.019$	$+0.001$	-0.001	-0.004	-0.12
Random flips, maximum over 40	$+0.029$	$+0.004$	$+0.004$	$+0.003$	0.53

HBS+Prior. Table 7 gives the effect of the LLM prior on the 69 pairs. From budget 1 onward, the proxy uses the measured priors. At budget 0, it models the prior for the k-th evaluation item of a pair as the K_0-shrunk mean of k ratings drawn around the benchmark's mean rating with the standard deviation of single ratings on the training benchmarks, 0.205; the mean rating is recovered by undoing the shrinkage of μ_b over the n_b items rated in the measurement run, and $|E_j|$ is read from each report. In the measurement run, the budget-1 error $e = |\mu_b - r_j|$ leaves the candidates $r_j \pm e$, whose budget-3 predictions have errors $|0.1 - e|$ and $0.1 + e$; these differ whenever $e > 0$, and no prior exceeded 0.89, so $\mu_b + 0.1$ was never clipped.

Accuracy of the priors. Across the seven test benchmarks, the priors order the benchmarks' mean success rates with Kendall's $\tau = 0.62$ and lower the mean absolute error from 0.22 for the constant 0.35 to 0.15, but they are compressed: regressing the priors on the mean rates gives a slope of 0.38. The prior lowers the expected Brier ALC on five of the seven benchmarks and on 51 of the 69 pairs; removing any one benchmark leaves an improvement of 0.0057 to 0.0088, and the jackknife standard error over benchmarks, 0.0033, is larger than the interval of Table 7 suggests. Table 8 collects the checks of Section 6.4. The priors were redrawn 20 times from their sampling distribution (the mean of n ratings with standard deviation 0.205, where n is the number of items rated in the measurement run); the order of the items was permuted 200 times for fixed ratings; and K_0, set to 5 before the measurement, was varied from 0 to 20. In the formative run of the final submission, predictions at budgets 1 and 3 are constant within five of the nine pairs and nearly constant within the other four, where only the corrections of Equation (4) vary, so both budgets approximately identify each pair's rate: their estimates of $r_j(1 - r_j)$ agree exactly for the five pairs and to within 0.004 for the others. Table 9 compares that run with the proxy's expectations on its pairs.

Table 7: HBS with the LLM benchmark prior on the 69 real pairs. The priors are fixed at the values measured in Section 5.3, so the 95% intervals, from resampling whole reports, cover label noise and the sampling of pairs, but neither the sampling error of the priors (Table 8) nor benchmarks outside the test pool.

Predictor	Brier ALC	Change from HBS
HBS	0.179	—
HBS+Prior, prior used from budget 1	0.175	-0.0044 $[-0.0052, -0.0035]$
HBS+Prior, prior used from budget 0 (final submission)	0.171	-0.0073 $[-0.0087, -0.0059]$
Each benchmark's rate known, $\alpha = 3$ (diagnostic)	0.162	-0.0166 $[-0.0185, -0.0149]$
Each benchmark's rate known, $\alpha = 6$ (diagnostic)	0.160	-0.0186 $[-0.0204, -0.0171]$

Baselines and tuning grids. The official baseline predicts the mean of the pair's revealed labels, and 0.5 without labels. The own-label Bayes predictor takes as its prior the empirical distribution of the success rates of all other pairs and predicts the posterior mean given the pair's own labels. The mixture prior adds to the pair prior of HBS, with weight w, a near-unsolvable component, a Beta distribution with mean m and strength k; we search $w \in \{0, 0.1, \ldots, 0.4\}$, $m \in \{0.02, 0.05, 0.1\}$ and $k \in \{5, 10, 30\}$. Re-tuning searches $\mu_0 \in \{0.3, 0.35, 0.4\}$, $\alpha \in \{2, 3, 4, 6\}$ and $\kappa_1 \in \{1, 3, 6\}$. Both select leave-one-report-out.

Table 8: Change in Brier ALC of HBS+Prior (prior from budget 0) relative to HBS on the 69 pairs under different checks.

Check	Change in Brier ALC
As submitted ($K_0 = 5$), 95% interval over reports	-0.0073 $[-0.0087, -0.0059]$
26 pairs with known roots	-0.0082
$K_0 = 0$, 2, 10, 20	-0.0078, -0.0077, -0.0069, -0.0061
Item order permuted 200 times (mean, SD)	-0.0073 (0.0001)
Priors redrawn 20 times (mean, SD)	-0.0073 (0.0008)

Table 9: The formative run of the final submission against the proxy's expectations for HBS and HBS+Prior on its nine pairs. The budget-0 expectation for HBS involves no labels and hence no label noise; the label-noise standard deviation of the Brier ALC is 0.007.

Budget	0	1	3	7	15	31	Brier ALC
Observed, final submission	0.190	0.192	0.183	0.163	0.163	0.151	0.174
Expected, HBS	0.224	0.201	0.181	0.167	0.160	0.156	0.180
Expected, HBS+Prior	0.194	0.182	0.173	0.165	0.159	0.155	0.171

F LLM ratings on the training benchmarks

Tables 10 and 11 and Figure 5a give the anchored benchmark-level ratings of Section 6.3 on the four held-out training benchmarks, and Figure 5b compares HBS+ItemLLM with the proxy at each budget.

Table 10: Budget-0 priors from anchored LLM ratings on the four held-out training benchmarks. Kendall's τ compares predicted and actual orderings; single-item rows shrink each item's own rating toward 0.35.

Budget-0 prior	Kendall τ	Brier
Constant 0.35 (HBS)	—	0.233
Qwen3-8B, mean rating of the benchmark	0.33	0.237
Qwen3.5-9B, mean rating of the benchmark	-0.67	0.256
Qwen3-32B, mean rating of the benchmark	0.00	0.285
Qwen3.8-27B, mean rating of the benchmark	0.67	0.217
Qwen3.8-27B, single item, shrinkage chosen in hindsight	—	0.220
Qwen3.8-27B, single item, shrinkage chosen LOBO	—	0.231
Oracle that knows each benchmark's rate	1.00	0.199

Table 11: Mean predicted success rate of each held-out training benchmark from anchored ratings, against its actual mean rate on the rated items.

	multi_swebench	researchcodebench	real_webagents	matharena
Actual	0.125	0.341	0.366	0.645
Qwen3-8B	0.206	0.108	0.378	0.340
Qwen3.5-9B	0.535	0.532	0.459	0.514
Qwen3-32B	0.330	0.384	0.700	0.210
Qwen3.8-27B	0.202	0.359	0.407	0.394

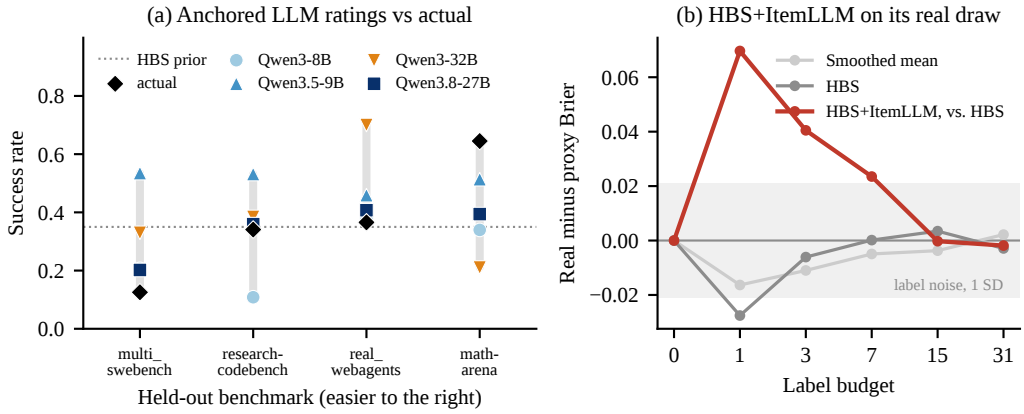


Figure 5: (a) Mean success rate predicted by anchored prompts for each held-out training benchmark (markers) against the actual rate (diamonds); dotted line: the budget-0 prediction of HBS. (b) Real minus proxy-expected Brier score per budget on each submission's own draw; HBS+ItemLLM is compared with the expectation for HBS.

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Answer: [Yes]

Justification: Proposition 1 states its assumption (a constant budget-0 prediction within the pair) and is followed by a complete proof (Section 5).

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